# Open and Challenging Problems in Kinematic Synthesis of Planar and Spatial Linkages

## Executive summary

Kinematic synthesis has a mature classical core—finite-position planar and spherical synthesis can be solved exactly up to *five* prescribed poses (a cornerstone of Ludwig Burmester[1]'s late-19th-century program), and the "maximum-control" planar four-bar path synthesis problem (nine precision points) has a complete enumerative solution via polynomial continuation. [2] Yet the field still contains multiple *open or persistently challenging* problems because practical design objectives are **not** captured by the classical "solve equalities at finitely many points/poses" paradigm: designers need inequality constraints (defect avoidance, transmission angle limits, collision clearance), robustness to tolerances, continuous-trajectory fidelity (not just finite points), topology/type choices, and scalable methods for higher-bar-count planar and spatial mechanisms. [3]

On the **model-based** side, the dominant bottleneck is that synthesis reduces to solving structured polynomial systems whose *raw algebraic complexity* (Bezout bounds, degrees, singularities/degeneracies) quickly becomes enormous; even when complete solution sets are computable, filtering them to "good mechanisms" becomes a semi-algebraic and often continuous-time feasibility problem. [4] On the **spatial** side, a sharp structural limitation is that common 1-DOF spatial four-bar topologies (e.g., RCCC) generally cannot realize more than three prescribed coupler poses exactly, so multi-pose spatial rigid-body guidance typically demands richer topologies (e.g., 6R families, multi-loop designs) and more sophisticated synthesis theories. [5]

On the **data-driven** side, the last few years produced large-scale synthetic datasets and promising retrieval/generative pipelines, but hard open issues remain: invariant representations of motion/path, multi-solution (one-to-many) inverse maps, topology generalization, trustworthy constraint satisfaction, and standardized evaluation. Representative developments include LINKS (100M planar 1-DOF mechanisms, 1.1B coupler curves), a ~3M sample dataset for 4-/6-/8-bar planar mechanisms with proposed metrics, and hybrid retrieval+optimization (LInK) with benchmark tasks such as "LINK-ABC" alphabet curves. [6]

The remainder of this report enumerates eight high-leverage problems and proposes concrete technical routines—symbolic–numeric elimination and homotopy continuation (plus certification), semidefinite/SOS relaxations for global polynomial optimization, mixed-integer formulations for discrete choices, topology-aware learning models, and hybrid pipelines that use learning to propose candidates while model-based methods enforce feasibility.

## Scope and synthesis paradigms

**Synthesis goals and task spaces.** Classical linkage synthesis problems are commonly categorized as function generation, path-point generation (coupler curve through precision points), and motion generation / rigid-body guidance (poses in SE(2) for planar, SE(3) for spatial). [7] Exact finite-position synthesis is possible only up to a task-dependent maximum: for planar four-bars, the classical Burmester five-pose rigid-body guidance problem is generically finite, while planar point-path synthesis is maximally constrained at nine precision points. [8]

**Model-based formulations.** The classical pipeline encodes kinematic constraints as polynomial equations (after appropriate coordinates, e.g., isotropic/complex coordinates in planar synthesis; dual quaternion / Plücker line coordinates in spatial). Complete solution strategies use elimination + continuation to obtain all isolated solutions, then filter geometric degeneracies and physical defects. [4] This is now tightly connected to **numerical algebraic geometry** toolchains (homotopy continuation and witness-set ideas) and software (e.g., Bertini, PHCpack, HomotopyContinuation.jl; solution certification via α-theory). [9]

**Data-driven formulations.** Learning-based synthesis treats the task as an inverse map from a curve/pose specification to mechanism parameters, often using generated simulation data; recent work highlights (i) representation of mechanisms/specifications in non-Euclidean spaces, (ii) mapping design specs to both type and dimensions, and (iii) generative architectures that produce multiple candidates rather than single regressions. [10] Large datasets (e.g., LINKS) enable contrastive retrieval and neural generative modeling at scale but also introduce dataset bias and evaluation challenges. [6]

**Comparative map of key open problems.** The table below summarizes eight problems treated in depth later.

| Problem ID | Problem (short name) | Planar / Spatial | Synthesis goal focus | Typical linkage families | Algebraic / computational signature | Data availability | Maturity of solutions | Estimated research effort |
|---|---|---|---|---|---|---|---|---|
| P1 | Defect-free, inequality-constrained planar | Planar | Path, motion, mixed (Alt-Burmester), function+ | 4-bar; extensions to 6-/8-bar | Polynomial equalities + *semi-algebraic* | High (synthetic), medium (ben | Strong for equality-only finite-point cases; weak for certified inequalities | High |

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|---|---|---|---|---|---|---|---|---|
| | synthesis | | motion | | constraints; continuous-time feasibility | ch) [6] | | |
| P2 | Scalable complete synthesis for 6-/8-bar planar mechanisms | Planar | Path/motion generation beyond 4-bar expressivity | Stephenson/Watt 6-bar; 8-bar | Very large sparse polynomial systems; many solution components | High for planar paths (LINKS) [11] | Partial (mostly optimization /heuristics); limited complete solvers | High |
| P3 | Certified continuous-trajectory approximation & robustness | Planar (extends spatial too) | Continuous path/motion (not just precision points) | 4-bar through multi-loop | Optimization over trajectories; robustness under tolerances; certification needed | Medium–high (synthetic) [12] | Early-stage standardization; limited certification | High |
| P4 | Practical curve | Planar (foundational | Coupler curve / curve | Kempe-style constru | Configuration- | Low–medium | Theory mature for existence; | High |

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|---|---|---|---|---|---|---|---|---|
| | drawing with low complexity & single-branch guarantees | ); some spatial analogs | drawing | ctions; factorization-based linkages | space components; universality with large constructions | (curve data sets) | design optimality largely open | |
| P5 | Minimal 1-DOF spatial mechanisms for multi-pose rigid-body guidance | Spatial | Rigid-body guidance in SE(3) | RCCC, CCCC dyads; beyond 4-bar | Pose constraints exceed topology capacity; high-degree elimination | Low–medium | Structural limits known; constructive synthesis beyond limits is open | High |
| P6 | Robust and complete spatial Burmester / dyad synthesis under degen | Spatial | Dyad synthesis; 4-bar assembly from dyads | CC, RC, RR dyads; RCCC/CCCC | Bivariate 12th-degree systems; resolvents up to degree 144; real- | Low–medium | Some robust methods exist; completeness+robustness remains hard | Medium–high |

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|---|---|---|---|---|---|---|---|---|
| | eracy | | | | solution filtering | | | |
| P7 | General spatial 6R synthesis for ≥5 poses with constraints | Spatial | Multi-pose rigid-body guidance; motion generation | 6R closed chains (beyond special subclasses) | Dual-quaternion polynomial factorization gives families for 4 poses; general case open | Low | Active research; limited to structured classes | High |
| P8 | Learning-based inverse synthesis with invariances, multi-solution output, and guarantees | Planar + Spatial | Path/motion/function; type+dimension | Broad families; learned embeddings | Multimodal inverse map; equivariances; constraint satisfaction | High planar; emerging spatial [13] | Rapid progress; guarantees/benchmarks still immature | Medium–high |

# Open problems in planar model-based synthesis

## P1 — Defect-free, inequality-constrained planar synthesis

1) **Problem and scope.**
Given a planar synthesis task (path generation, motion generation, function generation, or mixed Alt-Burmester), find mechanism dimensions (and possibly ground-pivot placements) such that (i) the mechanism interpolates or approximates the task, and (ii) it satisfies *inequality and "between-points" constraints*: no branch/order/circuit defects, adequate transmission angles, joint limits, collision clearances, and robustness to tolerances. The classical finite-position problems have enumerative solutions (e.g., nine-point path synthesis; families of Alt-Burmester problems), but these solutions can include many physically unacceptable candidates. [14]

2) **Why it is unsolved or challenging.**
The difficulty is structural: equality-constrained synthesis yields algebraic sets (solvable with homotopy continuation), but adding design requirements converts the problem into **semi-algebraic feasibility** and often **continuous-time verification** (properties must hold over ranges of motion, not only at precision points). Classical synthesis literature explicitly flags the need to check acceptable motion between precision points, detect branch/circuit defects, evaluate transmission characteristics, and assess sensitivity to dimensional variation after solving precision-point synthesis. [15] Computationally, even "just" enumerating equality solutions can involve very large Bezout counts before reduction (e.g., the nine-point four-bar formulation has a 2-homogeneous Bezout number of 286,720; practical continuation tracks 1,442 paths after reduction). [16] Turning these enumerations into certified defect-free designs adds another layer of difficulty.

3) **Prior work and key references.**
Finite-point completeness and structure: - Complete nine-point planar four-bar path synthesis via polynomial continuation, with explicit treatment of degenerate solutions and practical defects (branch/order defects discussed in examples). [17]
- Alt-Burmester family: dimensions, degrees, and which mixed (M poses, N points) problems are zero- vs one-dimensional; explicit emphasis on post-filtering for defects and sensitivity. [18]
Task feasibility constraints: - A four-bar coupler curve intersects a circle/line at most six times; hence, feasibility constraints exist even at the level of choosing precision points. [19]
- Longstanding practical gap: selecting precision points to approximate a desired continuous curve (e.g., ellipse-like shapes) is not principled and interacts with transmission-angle quality. [20]

4) **Concrete technical routines and research directions.**
A pragmatic research program is to treat inequality constraints as *first-class* objects and build a certified pipeline:

**Routine A: Enumerate → filter → certify (semi-algebraic screening).** - Step 1: Solve equality synthesis completely using numerical algebraic geometry with structure-exploiting homotopies (multihomogeneous, parameter homotopy, regeneration). [21]
- Step 2: Filter for real solutions and geometric non-degeneracy (exclude known degenerate families explicitly identified in the polynomial formulation). [22]
- Step 3: Evaluate discrete design inequalities at finitely many samples along the motion parameter and reject quickly.
- Step 4: *Certify* remaining candidates with: - certified solution checks for the equality constraints (α-theory via Jonathan D. Hauenstein[23]–Frank Sottile[24]'s alphaCertified) for numerical correctness of roots; [25]
- interval-based motion verification for continuous constraints (e.g., transmission angle bounds) by enclosing the motion curve and verifying inequalities over the parameter interval (the general "interval arithmetic" + certification paradigm is standard in certified numerical computation). [26]

**Routine B: Polynomial optimization with SOS/SDP relaxations.** - Formulate approximate synthesis as minimizing a polynomial (or rational, then polynomialized) objective, subject to polynomial equalities/inequalities capturing closure + design constraints.
- Apply the Lasserre moment/SOS hierarchy to obtain converging lower bounds and, when tight, global certificates. [27]
- Use Parrilo-style SOS relaxations for semialgebraic constraints and infeasibility certificates. [28]
This is particularly compelling when constraints can be written as polynomial inequalities in a finite parameterization (e.g., enforcing angle cosines/sines via auxiliary variables), but will require careful sparsity exploitation and problem-specific hierarchies.

**Routine C: Discrete choices via mixed-integer convex relaxations.** - Encode topology/assembly-mode selection or collision-avoidance disjunctions as mixed-integer constraints, and relax nonconvex rotation constraints through semidefinite descriptions of the convex hull of rotation matrices where appropriate (especially for SE(3)-related subproblems or local pose constraints). [29]
- Use mixed-integer convex programming as a global search scaffold, then refine with local nonlinear optimization.

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility is highest when: - inequality constraints are "local" (checkable at finite samples) and monotonicity/regularity can be established to make interval

verification tractable; - the equality solution sets are zero-dimensional (finite) so enumeration is realistic (as in classic maximal-precision problems). [8]

Expected difficulties: - certification over continuous motion may require bounding configuration-space branches and avoiding singularities (a central practical issue in synthesis examples). [30]
- SOS relaxations can be large and may need chordal sparsity or problem decomposition to scale. [31]

Evaluation experiments: - Benchmark on Alt-Burmester problem instances where full equality solution sets are known (dimension/degree tabulated) and compare: (i) percent of equality solutions that survive defect-free constraints, (ii) runtime, (iii) certification success rate. [18]
- Use near-degenerate point sets (e.g., many points near a circle) to test robustness; these are explicitly identified as difficult in complete-solution studies. [22]

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**P2 — Scalable complete synthesis for 6-/8-bar planar mechanisms**

1) **Problem and scope.**
Develop "complete" (or at least coverage-guaranteed) synthesis for 1-DOF planar six- and eight-bar mechanisms that can realize richer paths/motions than four-bars, with either finite precision targets or continuous approximations. This includes topology selection (Stephenson/Watt six-bars, etc.) and dimensional synthesis under path/motion objectives.

2) **Why it is unsolved or challenging.**
Even for four-bars, complete precision-point synthesis can require sophisticated degree reduction and continuation to manage huge Bezout numbers (e.g., reduction from millions of potential solutions to 1,442 tracked paths in nine-point synthesis). [16] For higher-bar mechanisms, polynomial systems typically become larger, more sparse, and more singular, with additional loop-closure constraints and more frequent positive-dimensional solution components. Classical theory emphasizes that exact matching of arbitrary continuous tasks is impossible and motivates finite-point approximations—but higher-bar mechanisms are precisely used because practical tasks exceed four-bar expressivity, increasing the need for large-scale approximate synthesis. [15]

3) **Prior work and key references.**
- Large-scale simulation datasets now include multi-bar mechanisms, enabling empirical and learning-based exploration: LINKS covers planar 1-DOF mechanisms up to ~20 joints and provides normalized paths and curated subsets to reduce bias. [11]
- A dedicated dataset paper reports ~3 million planar 4-/6-/8-bar mechanisms with

open/closed coupler curves and proposes standardized evaluation metrics. [32]
- Classical four-bar maximal-precision theory provides a template for what "complete" means (enumerate all solutions, then filter), but scaling that template to larger linkage graphs is nontrivial. [8]

4) **Concrete technical routines and research directions.**
**Routine A: Structure-exploiting numerical algebraic geometry for multi-loop planar systems.** - Derive polynomial systems in a coordinate system that minimizes degrees (analogous to the multihomogeneous reductions used in four-bar synthesis). [30]
- Compute mixed volumes (BKK bounds) and design multihomogeneous groupings to reduce path counts; then use regeneration-style strategies to solve equation-by-equation. (Regeneration is commonly cited as a powerful path count reduction strategy in mechanism polynomial systems.) [33]
- Track solution sets across parameter homotopies (task changes) to amortize cost across benchmarks, as done in classical continuation-based kinematics solving. [30]

**Routine B: Elimination templates / eigenvalue solvers for planar synthesis subproblems.** - Borrow the "elimination template → generalized eigenproblem" paradigm used for Stewart–Gough forward kinematics: construct sparse elimination templates tailored to planar linkage equations and extract solutions via linear algebra. [34]
- Key research step: identify minimal monomial bases and stable pivoting strategies (often problem-specific) to keep templates small and numerically stable.

**Routine C: Hybrid retrieval+optimize at topology scale.** - Use contrastive retrieval to propose candidate topologies and initial geometries from large datasets, then refine via constrained optimization. LInK is an explicit demonstration of this "retrieve from millions → warm-start hierarchical optimization" pattern for planar path synthesis. [35]
- Extend retrieval embeddings to include topology (graph) and kinematic feasibility regions, not only curve similarity.

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility hinges on exploiting sparsity and decomposition—without it, complete solvers will be dominated by path counts and singularities. Expected difficulties include: - handling positive-dimensional solution sets (curves/surfaces of solutions) that arise when tasks are under-specified or when additional degrees of freedom appear; [18]
- reconciling dataset-driven candidate selection with strict constraint satisfaction.

Evaluation experiments: - Define a benchmark suite of target curves (simple, moderate, complex), measure: success rate, best-achieved error, diversity of solutions, and constraint violations, using the standardized metrics proposed in the ~3M dataset paper as a baseline. [32]

- Compare complete/coverage-seeking solvers against pure optimization baselines and retrieval-only baselines using LINKS-normalized paths. [36]

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**P3 — Certified continuous-trajectory approximation and robustness**

1) **Problem and scope.**
Move beyond "fit N precision points/poses" toward **continuous** path/motion objectives with explicit error bounds, while incorporating robustness to manufacturing tolerances and measurement noise. This applies to planar (and spatial) linkages, but planar four-bar synthesis exposes the issue sharply: even when you can compute all exact-interpolation designs, the deeper design question is how to choose tasks/points to approximate desired continuous curves without creating infeasibility or poor transmission. [37]

2) **Why it is unsolved or challenging.**
Continuous-trajectory matching introduces functional objectives (integrals, supremum norms, Fréchet-type distances) and "for all time" constraints, which are much harder than finite constraints. Even the classical literature notes that picking precision points for curve approximation is not principled and trades off against transmission quality. [20] Robustness adds uncertainty propagation across kinematics, which is nonlinear and can be discontinuous across branches.

3) **Prior work and key references.**
- Explicit statements of the precision-point limitation and the need for post-synthesis checks (motion between points; sensitivity to variation). [18]
- Recognition of feasibility constraints induced by coupler-curve algebraic structure (e.g., limits on intersections with lines/circles constrain point selection). [38]
- Emerging dataset work explicitly calls out the lack of standardized outcome metrics and proposes multiple metrics for comparing learning-based synthesis outcomes. [32]

4) **Concrete technical routines and research directions.**
**Routine A: Adaptive precision-point selection with certification loops.** - Start with an initial discretization of the target curve (Chebyshev-like spacing where appropriate), solve synthesis, then compute continuous error and adaptively add points in regions of maximum deviation.
- Use interval-based enclosure of the generated coupler curve segment to certify inequality constraints and bound approximation error.

**Routine B: Polynomialized continuous objectives for SOS bounds.** - Approximate the continuous objective by polynomial quadrature (or represent certain curve families exactly as polynomials/trigonometric polynomials), reducing the problem to polynomial optimization on a compact set.

- Apply Lasserre/Parrilo hierarchies to obtain lower bounds and feasibility certificates for candidate designs. [31]

**Routine C: Robust design via uncertainty quantification (UQ).** - Model tolerances as random variables; propagate through kinematics via polynomial chaos or Monte Carlo on candidate mechanisms.
- Optimize a risk-aware functional (e.g., expected path error + tail risk), and compare to nominal objectives.

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility is best for: - restricted motion ranges where branch switching is excluded by construction;
- target curves with known parameterizations that make continuous error measurable and differentiable.

Difficulties: - ensuring the chosen branch/order of motion is consistent (a known defect class in synthesis examples); [30]
- scaling SOS relaxations to high-dimensional design variables.

Evaluation experiments: - For each target curve, report (i) maximum deviation over the cycle, (ii) robustness under perturbations, (iii) constraint violation probability.
- Use the datasets' proposed metrics for comparability, but augment with robustness/continuous-error metrics. [32]

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**P4 — Practical curve drawing with low complexity and single-branch guarantees**

1) **Problem and scope.**
Given a target plane curve (algebraic, parametric, or approximated), design a planar linkage that draws it with **small link count**, reasonable conditioning (size, joint angles), and ideally a **single connected configuration component** that traces only the intended curve segment (no spurious branches). This includes classical "curve drawing" goals and also modern mechanism design tasks where coupler curves must match complex paths.

2) **Why it is unsolved or challenging.**
Existence theorems show remarkable generality but do not yield *useful* designs. Alfred Bray Kempe[39]'s 1876 construction describes plane algebraic curves by linkwork, but it is intrinsically complex. [40] Modern mathematics proves universality of planar linkage configuration spaces (moduli spaces), implying that extremely complex behavior is representable—again without giving minimality or practicality. [41] Contemporary engineering work on drawing algebraic curves explicitly notes that Kempe-style linkages tend to be "exceedingly complex," motivating alternative algorithms that work for restricted curve classes. [42] The underlying optimization is combinatorial (graph structure) plus nonlinear geometry;

moreover, configuration spaces can have multiple components, and ensuring "single-branch" drawing is a topological problem as much as an algebraic one. [43]

3) **Prior work and key references.**
- Original curve-drawing linkage method (Kempe, 1876). [40]
- Universality theorems for planar linkage configuration spaces (Kapovich–Millson). [44]
- Engineering synthesis of linkages to draw plane algebraic curves, explicitly discussing Kempe complexity and proposing constructive methods. [42]
- Algorithmic simplification for *parametric* motions via noncommutative factorization methods ("linkages following a prescribed motion"). [45]

4) **Concrete technical routines and research directions.**
**Routine A: Constrained factorization pipeline for parametric motions.** - Restrict to rational/parametric curve or rigid-motion families where motion-polynomial factorization applies; generate candidate linkages by factorization; then post-optimize link lengths/joint placements for conditioning and size. [46]
- Add constraints during factor selection (e.g., avoid near-parallel axes, enforce bounded link lengths) to produce mechanically reasonable designs.

**Routine B: Minimality as mixed-integer + continuous optimization.** - Introduce a library of linkage "gadgets" (additor/multiplicator/translator-like primitives) and solve for the smallest-cost combination that approximates a target curve within tolerance.
- Encode gadget selection as integer variables; encode geometry as continuous variables; solve with branch-and-bound plus strong relaxations (SOS/SDP for polynomial constraints where applicable). [47]

**Routine C: Topological component control.** - Compute or approximate connected components of configuration spaces for candidate designs (numerical algebraic geometry witness sets + sampling), and enforce that the intended drawing corresponds to a single component. [48]
- Develop heuristics/criteria to avoid designs where the target curve is drawn only on a small component while other components cause spurious motion (a known issue in Kempe-related constructions). [49]

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility is strongest when the target is parametric and admits factorization-based construction; it is weakest for arbitrary implicit curves where Kempe universality is the only guarantee. [50]

Expected difficulties: - component computations can be expensive;
- optimizing for minimality is likely computationally hard;
- mechanical conditioning (link lengths, joint angles) can conflict with exactness.

Evaluation: - Compare link counts, envelope size, condition numbers, and single-component success across a fixed set of curves (conics, cubics, epitrochoids, and selected "signature-like" curves), with both exact and approximate targets. [51]

## Open problems in spatial model-based synthesis

**P5 — Minimal 1-DOF spatial mechanisms for multi-pose rigid-body guidance**

1) **Problem and scope.**
Design low-complexity spatial mechanisms (preferably 1-DOF, closed-loop) that guide a rigid body through multiple prescribed poses (position + orientation). The key scope distinction is that "spatial four-bar" linkages exist in multiple joint-type families (e.g., CCCC, RCCC), and the synthesis target is typically rigid-body guidance, not only point-path tracing.

2) **Why it is unsolved or challenging.**
A central difficulty is *capability mismatch*: in a detailed treatment of the spatial Burmester problem for spatial four-bars, it is shown that CC dyads can meet up to five poses, RC dyads only up to three, and that an RCCC spatial four-bar (the spatial counterpart of planar/spherical four-bars in that formulation) can visit up to **three** given poses—so there are **no exact solutions for RCCC synthesis when m > 2** (i.e., beyond three poses). [5] This creates a hard "minimal topology" question: what is the smallest 1-DOF spatial linkage family that can realize 4–5 poses, and how can it be synthesized reliably?

3) **Prior work and key references.**
- Spatial Burmester / spatial four-bar synthesis (CCCC and RCCC types) with explicit solvability limits and equation structure. [5]
- Continuation methods as a pathway to "all solutions (real and complex)" in spatial mechanism synthesis (noting that many solutions are complex and must be filtered). [52]

4) **Concrete technical routines and research directions.**
**Routine A: "Topology ladder" synthesis search.** - Start from dyadic building blocks (CC, RC, RR) and incrementally add constraints/links until the pose count becomes feasible (4–5 poses).
- At each rung, formulate polynomial systems in dual-quaternion or line-coordinate representations; solve with NAG; then enforce 1-DOF closure by adding mobility constraints (e.g., Jacobian rank conditions) as additional equations/inequalities.

**Routine B: Certified feasibility frontiers for pose counts.** - Use dimension counting and numerical algebraic geometry to empirically map "pose-count feasibility regions" for each topology class, including the existence of positive-dimensional solution families. [53]
- Deliverable: a benchmark atlas specifying for each topology the maximal exact

pose count under generic conditions, plus examples of special geometries that exceed generic bounds.

**Routine C: Hybrid elimination-template solvers for spatial synthesis substeps.**
- Where the formulation yields high-degree eliminations (resolvents), substitute elimination-template eigenvalue solvers (as used in modern Stewart–Gough forward kinematics solvers) for numerical stability and speed. [34]

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility depends on carefully defining "minimal" (fewest joints? shortest loop count? single actuator?) and on selecting pose sets that are physically meaningful (not arbitrary in SE(3)). Expected difficulties include: - high sensitivity near singular configurations;
- multiplicity of solutions and assembly modes;
- ensuring 1-DOF behavior after synthesis.

Evaluation: - Use two benchmark classes: (i) random pose sets in SE(3) with controlled spacing, (ii) application-derived pose sequences (pick-and-place, rehabilitation motion capture) as in recent spatial-learning demonstrations. [54]

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**P6 — Robust and complete spatial Burmester/dyad synthesis under degeneracy**

1) **Problem and scope.**
Compute *all real, physically meaningful* solutions to spatial dyad synthesis problems (especially CC dyads for five poses) and assemble them into candidate spatial four-bars (CCCC, RCCC where applicable), while robustly handling degenerate cases: none or infinitely many solutions, near-singular inputs, and numerical instability.

2) **Why it is unsolved or challenging.**
Spatial dyad synthesis can lead to very high-degree algebraic elimination: one formulation produces four bivariate 12th-degree polynomials, and dialytic elimination can yield a resolvent degree as high as 144, which is numerically delicate. [5] Additionally, it is explicitly noted that compared with planar/spherical Burmester problems, spatial rigid-body guidance synthesis has received much less attention and few works exist for spatial four-bars, so robust "standard" toolchains are less established. [5]

3) **Prior work and key references.**
- A semigraphical approach that "filters out complex solutions" to ease solving and handles special cases of none/infinite solutions, while still acknowledging arbitrariness in parameter choices. [5]
- General certified numerical methods and software for polynomial systems (Bertini; PHCpack; HomotopyContinuation.jl) and certification (α-theory). [9]

4) **Concrete technical routines and research directions.**
**Routine A: Complete-and-certified homotopy solving for spatial dyads.** - Use parameter homotopies from generic instances to specific pose sets to compute all isolated complex solutions; then test realness and physical acceptability. [55]
- Apply certified tracking / a posteriori certification to avoid missing solutions near singular endpoints (which are explicitly observed even in planar problems and are common in spatial ones). [56]

**Routine B: Real-solution filtering as geometry, not heuristics.** - Integrate "real algebraic geometry" filtering steps (e.g., sign conditions, interval isolation) early rather than as a final step.
- When semigraphical filtering is used, formalize it as a deterministic projection + real-root isolation procedure to remove arbitrariness.

**Routine C: Resilient elimination strategies.** - Replace explicit degree-144 univariate elimination with elimination templates + eigenvalue extraction, which is designed to be numerically robust and relies on linear algebra operations rather than explicit high-degree polynomial root finding. [57]

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility is good for "batch solving" many instances if parameter homotopy reuse is exploited. Expected difficulties: near-degeneracy, clustered roots, and ensuring non-duplication of geometrically equivalent solutions (symmetry).

Evaluation: - Construct pose datasets with controlled degeneracy (e.g., poses near a screw motion; nearly coplanar axes).
- Score solvers by completeness (number of certified solutions found), robustness (failure rate under noise), and physical filtering accuracy.

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**P7 — General spatial 6R synthesis for five or more poses with constraints**

1) **Problem and scope.**
Design spatial 6R closed chains (and related overconstrained families) that realize multi-pose rigid-body guidance. A key sub-scope is moving from special structured subclasses (e.g., angle-symmetric) to general 6R synthesis with constraints on joint axes, link lengths, and avoidance of singularities.

2) **Why it is unsolved or challenging.**
For inverse kinematics (related but not identical), structured polynomial formulations already exist with high algebraic complexity (e.g., an 11-equation formulation with total degree 1024 but generic solution count 16 for the general 6R inverse position problem). [58] Synthesis adds another layer: the mechanism parameters themselves become unknowns alongside configuration variables. While motion-polynomial factorization over dual quaternions provides constructive

synthesis for certain four-pose 6R families, it typically yields either no solution or a one-parameter family, and selecting “good” solutions is nontrivial. [59]

3) **Prior work and key references.**
- Generic-case polynomial methodology for the 6R inverse position problem, demonstrating 16 generic solutions. [58]
- Four-pose synthesis of angle-symmetric 6R linkages via dual-quaternion motion-polynomial factorization, explicitly producing one-parameter families and discussing selection strategies. [59]

4) **Concrete technical routines and research directions.**
**Routine A: Pose-to-motion fitting, then factorization with constraints.** - Fit a rational SE(3) motion (dual-quaternion polynomial) to pose sequences (exact interpolation for small pose counts; least-squares for larger), then factor into revolute factors subject to mechanical constraints. [60]
- New research contribution: constrained factorization where admissible factors are restricted (axis bounds, limited twist), turning factor selection into a constrained algebraic problem.

**Routine B: Hybrid symbolic–numeric synthesis with continuation.** - Use factorization to obtain a low-dimensional solution family; then impose additional constraints (e.g., link length bounds, singularity margins) as extra equations/inequalities and solve via continuation + optimization.

**Routine C: Integrate robust global IK tool ideas.** - Incorporate convex relaxations for rotations (via convex hull spectrahedra for SO(n)) inside outer optimization loops to reduce sensitivity to local minima in axis-fitting subproblems. [61]

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.**
Feasibility is highest for structured families where the algebra provides a reduced parameterization (e.g., 4-pose special classes). Extending to ≥5 poses likely requires either higher-degree motion models, additional joints, or accepting approximation.

Evaluation: - Use controlled benchmarks: 4-pose tasks where structured solutions exist (compare to factorization baseline) and 5-pose tasks where feasibility is uncertain (report approximation and constraint satisfaction).
- Report diversity of solutions (family coverage) and robustness (sensitivity to pose perturbations).

## Open problems in data-driven and hybrid synthesis

**P8 — Learning-based inverse synthesis with invariances, multi-solution output, and guarantees**

1) **Problem and scope.**
Learn mappings from design specifications (curves, poses, mixed constraints) to

mechanisms (type + dimensions) for both planar and spatial linkages, while producing **multiple** valid candidates and respecting invariances (translation/rotation/scale; curve reparameterization; symmetries/cognates). This includes pure learning, retrieval-based, and hybrid pipelines.

2) **Why it is unsolved or challenging.**
Inverse synthesis is fundamentally **ill-posed**: many mechanisms can realize similar paths; conversely, small param changes can cause large motion differences; feasibility regions in design space can be disconnected. [62] Learning models also face non-Euclidean representations: mechanisms are naturally graphs, while motion/path outputs live in geometric spaces; a dedicated research perspective identifies representation, mapping (including type+dimension), and architecture as three key frontiers. [63] Even in four-bar path synthesis, multiple curve representations compete (points, Fourier, wavelets, images) and there is no universally best representation—recent work proposes using a VAE latent space as a unifying invariant representation and uses k-NN in latent space to generate multiple mechanisms. [64]

3) **Prior work and key references.**
Datasets and benchmarks: - LINKS dataset: 100 million planar 1-DOF mechanisms and 1.1 billion coupler curves, including normalized paths and curated sets to reduce path bias. [11]
- ~3M planar 4-/6-/8-bar dataset with open/closed coupler curves, motivated by "scarcity of high-quality datasets," proposing six outcome metrics. [32]
- LInK: contrastive learning on ~10 million mechanisms for fast retrieval, then hierarchical optimization warm-start; introduces a harder benchmark (LINK-ABC) based on capital alphabets and reports large improvements vs prior methods. [35]

Learning architectures and representations: - VAE-based invariant latent representations for coupler curves plus k-NN retrieval to produce multiple candidate mechanisms. [64]
- Generative deep models for four-bar synthesis (e.g., conditional GAN frameworks designed to generate diverse mechanisms satisfying kinematic and quasi-static conditions). [65]
- Early neural-network approaches to mechanism dimensional synthesis (trajectory shape synthesis) already identified optimization slowness and initialization dependence as motivations for learning. [66]

Spatial learning indicators: - An NSF record describes deep learning for spatial RSCR path synthesis and notes a database of >5 million paths; the full text is scheduled to become publicly available on April 1, 2026 (after the current date of March 7, 2026), so only abstract-level details are accessible in this report. [54]

4) **Concrete technical routines and research directions.**
**Routine A: Equivariant, multi-modal representations (curve + mechanism graph).** - Use geometric deep learning / graph neural networks for mechanism

topology; encode paths with transformation-invariant pipelines (normalization layers as in VAE coupler-curve work; contrastive objectives as in LInK). [67]
- Treat curve representations as *multiple views* (points/Fourier/wavelets/images) and enforce cross-view consistency in latent space (explicitly motivated by the "unifying latent representation" result). [64]

**Routine B: Probabilistic inverse maps with diversity and validity.** - Replace single regression with conditional generative models: conditional diffusion or normalizing flows, or VAE/GAN variants, trained to output sets of plausible mechanisms. (The need for "multiple candidate mechanisms" is explicit in the mechanism-design deep learning perspective and in generative work that emphasizes multiple distinct outputs.) [68]
- Build explicit *validity discriminators* trained on physical feasibility labels (closures, joint limits) and use them as guidance/filters.

**Routine C: Hybrid "learned proposal → model-based enforcement."** - Adopt LInK-style retrieval+optimization as a general template: (i) retrieve candidates quickly from a massive dataset; (ii) warm-start a robust local solver; (iii) enforce constraints with model-based checks. [35]
- Extend with homotopy continuation when constraints define polynomial systems: learned models propose near-solutions; continuation refines to certified solutions (α-theory as the final certificate). [69]

**Routine D: Dataset generation with controllable bias and coverage metrics.** - Follow LINKS in providing normalized paths and curated subsets; formalize "coverage" metrics (e.g., embedding-space uniformity, novelty under topology constraints). [11]
- For spatial datasets, mirror the planar strategy but include pose-space diversity, axis distributions, and singularity labeling; the RSCR database effort points to feasibility of multi-million-scale spatial datasets. [54]

5) **Feasibility, assumptions, expected difficulties, and evaluation experiments.** Feasibility is favorable in planar settings because datasets are already large-scale. [70] The main expected difficulties are: - guaranteeing constraint satisfaction (learning models may be approximate);
- out-of-distribution generalization to new topology classes or rare feasible regions (noted by LInK's creation of more challenging benchmarks where feasible space is "tiny"). [35]

Evaluation experiments should include: - standardized metric suites (start with the six metrics proposed alongside the ~3M dataset) plus constraint-violation metrics and diversity metrics; [32]
- cross-benchmark generalization: train on LINKS-like distributions and evaluate on LINK-ABC and classical "hard" curves (near-circular degeneracy cases, straight-line generator targets). [71]

- “physics audit” experiments: after prediction, run a certified closure and defect analysis pipeline; score the fraction of outputs that are certifiably valid.

## Benchmarks, datasets, and evaluation protocols

**Existing high-value datasets/benchmarks (suggested as defaults).**

| Resource | What it covers | Scale and notable features | Access / notes |
|---|---|---|---|
| LINKS dataset | Planar 1-DOF mechanisms (varied topologies) for path generation | 100M mechanisms; 1.1B coupler curves; provides simulations, normalized paths, curated set to reduce bias. [11] | 공개 PDF and code references described in the paper. [11] |
| ~3M planar 4-/6-/8-bar dataset | Planar path synthesis oriented toward ML evaluation | Nearly 3M samples; includes open and closed coupler curves; proposes six key evaluation metrics; demonstrates VAE + kNN use. [32] | Metadata and abstract accessible via NSF PAR record. [32] |
| LInK benchmark suite | Planar path synthesis with discrete + continuous variables | Trains on ~10M mechanisms; introduces harder benchmark LINK-ABC (alphabet trajectories) and reports strong speed/error improvements using retrieval + optimization. [35] | Accessible as arXiv HTML; includes benchmark discussion. [35] |
| Classical “complete-solution” benchmarks | Four-bar nine-point path synthesis; Alt-Burmester families | Complete equality solutions available; includes known hard near-degenerate cases and defect discussions. [30] | Use as gold standards for solver completeness and defect filtering. [30] |
| Spatial RSCR ML record | Spatial 1-DOF RSCR path synthesis | Abstract reports >5M paths and multiple DL models; full text embargoed until April 1, 2026. [54] | Plan around delayed full-text availability. [54] |

**Recommended evaluation metrics (cross-cutting).**
A robust evaluation suite should separate: (i) *geometric fidelity*, (ii) *constraint*

*satisfaction*, (iii) *solution diversity*, (iv) *computational cost*, and (v) *robustness*. The ~3M dataset paper explicitly argues that standardized metrics are missing and proposes a set of metrics; those should be treated as a baseline for comparability. [32] Augment with: - defect indicators (branch/order/circuit) and transmission-angle summaries (min/percentile over cycle), motivated by classical post-check requirements; [8]
- robustness under perturbations (Monte Carlo or certified bounds);
- diversity/coverage (e.g., embedding-space coverage, topology variety), motivated by large-dataset curation concerns. [36]

**Suggested visual taxonomy (Mermaid).**

```
flowchart TB
  A[Kinematic synthesis tasks] --> B[Function generation]
  A --> C[Path generation]
  A --> D[Motion generation / rigid-body guidance]

  C --> C1[Finite precision points]
  C --> C2[Continuous curve approximation]

  D --> D1[Planar SE(2)]
  D --> D2[Spatial SE(3)]
  D --> D3[Spherical]

  A --> E[Approach families]
  E --> E1[Model-based: algebraic / geometric]
  E --> E2[Data-driven: learned inverse maps]
  E --> E3[Hybrid: retrieve/learn + optimize/verify]

  E1 --> F[Polynomial systems]
  F --> F1[Elimination / Groebner / resultants]
  F --> F2[Homotopy continuation + certification]
  F --> F3[Polynomial optimization (SOS/SDP)]
```

## Research roadmap and feasibility

A plausible high-impact roadmap is to treat synthesis as a **stack** where each layer has a research-grade “done condition”:

1) **Complete equality solving (where possible) + certification.**
Leverage mature numerical algebraic geometry toolchains and certification (α-theory) to ensure computed solution sets are correct in maximal-precision problems, then use parameter homotopies to amortize across many instances. [72]

2) **Semi-algebraic filtering and continuous-time verification.**
Turn defect-free and geometric constraints into explicit verification modules; use interval methods and/or SOS relaxations where polynomial structure permits. This

directly addresses the long-recognized gap between “solutions exist” and “solutions are usable.” [73]

3) **Scaling to higher topologies via decomposition + learning.**
Exploit large datasets (LINKS) and retrieval+optimization (LInK) so that higher-topology design becomes feasible in practice, while model-based solvers remain the authority for correctness and constraints. [36]

4) **Spatial synthesis frontiers: topology limits and constructive synthesis beyond RCCC.**
Use the explicit solvability limits for spatial four-bar families as constraints on what “minimal” can mean, and develop topology ladders and constrained factorization methods to reach 4–5 pose guidance targets. [74]

**Timeline visual aid suggestion (Mermaid).**

```
timeline
  title Milestones motivating current open problems
  1876 : Kempe publishes general curve drawing by linkwork (existence, very complex)
  1888 : Burmester formalizes finite-position planar synthesis (Burmester problem lineage)
  1954 : Freudenstein introduces analytical 4-link function generation formulation
  1990 : Polynomial continuation methods enter kinematic synthesis practice
  1992 : Complete nine-point planar four-bar path synthesis via continuation (enumeration + defects)
  2002 : Universality theorems for planar linkage configuration spaces (modern proof framework)
  2022 : LINKS dataset (100M planar mechanisms) enables large-scale learning/retrieval
  2024-2025 : VAE/contrastive/generative pipelines + new datasets/benchmarks for ML synthesis
```

**Final feasibility note.**
The near-term “most feasible” technical wins are in hybrid pipelines that (i) use large datasets for candidate generation and diversity, and (ii) use model-based, possibly certified checks to guarantee closure, realness, and constraint satisfaction. This direction is strongly supported by the coexistence of (a) mature continuation/certification tools in polynomial system solving, and (b) rapidly expanding datasets and retrieval/generative methods in mechanism synthesis. [75]

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